



ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Mid Term-I (Session: 2025-26)

Roll No.:

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Course Name: B.Tech

Subject Name: Basics of Electronics Engineering

Subject Code:25EC101

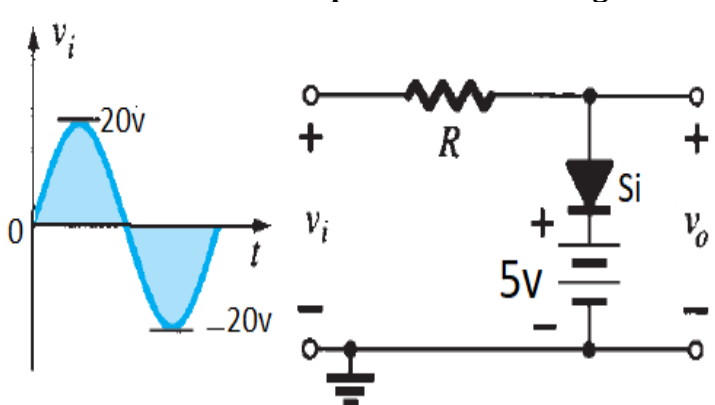
Max. Marks: 15

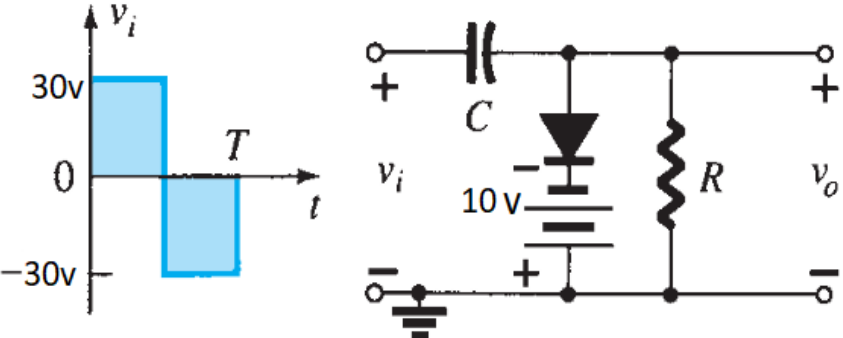
Time: 75 Minutes

Semester:1

Note:

1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Maximum Knowledge Level: K2)				
Q. No.	Question	CO	KL	PI
1. Attempt All Questions		(1 x 5)		
(a)	Define barrier potential of a P-N junction diode.	CO1	K2	2.4.4
(b)	Define Peak inverse voltage (PIV). Write the PIV rating of Centre tap transformer type full wave rectifier.	CO1	K2	2.3.1
(c)	Define ripple factor and write the values for the half wave and full wave rectifiers.	CO1	K2	2.3.1
(d)	BJT is bipolar or unipolar? Justify your answer.	CO2	K2	15.1.1
(e)	Why the size of collector is larger than emitter?	CO2	K2	1.3.1
SECTION-B (Minimum Knowledge Level: K2)				
2. Attempt Any two Question		(Same K Levels) (2 x 2)		
(a)	Draw the V-I characteristics of p-n junction diode. Also define breakdown voltage.	CO1	K2	16.1.1
(b)	Explain the working of Centre tap transformer type Full wave rectifier.	CO1	K2	5.1.2
(c)	Explain the application of Zener diode as a voltage regulator.	CO1	K2	16.1.1
SECTION-C (Minimum Knowledge Level: K3)				
3. Attempt Any one Question		(Same K Levels) (3 x 1)		
(a)	Determine and draw the output of the network given below. 	CO1	K3	2.1.3

(b)	Determine and draw the output of the network given below. 	CO1	K3	2.1.3
4. Attempt Any one Question (Same K Levels) (3 x 1)				
(a)	Explain the working of N-P-N transistor. If the value of α is 0.988 then determine the value of I_C ($I_E=10.2$milliampere).	CO2	K3	15.1.1
(b)	Draw and explain the input and output characteristics of Common base configuration. If $I_C= 10$milliampere & $I_E = 10.2$milliampere then determine the value of α.	CO2	K3	15.1.1

CO Course Outcomes mapped with respective question

KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create